

Lesson 6.18: Careers in Food Systems, and the Unit 6 Review and Test

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.4 Careers
Grade: 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (Day 1 careers; Day 2 review game and the Unit 6 test) **Room:** Either (classroom or FACS lab; no cooking either day)

Standards

- NYS module line: NYS FSP 6. Career Pathways a) Explain roles and functions of individuals engaged in food industry careers
- NYS module line: NYS FSP 6. Career Pathways b) Investigate education, training requirements, and opportunities for career paths in global food industry
- NYS module line: NYS FSP 6. Career Pathways c) Assess personal employability skills for careers in the global foods industry and evaluate personal suitability for said careers
- CTE theme line: CTE CCO 4. Career Plans a) Examine potential career choices to determine knowledge, skills, and abilities associated with each
- CTE theme line: CTE CCO 7. Career Pathways c) Research ways social, economic, and technological changes have led to adaptations in work and community environments and expectations
- CTE theme line: CTE SUS 4. Careers Related to Sustainability a) Investigate knowledge, skills, and practices needed for careers related to sustainability or for employment in settings that utilize sustainable practices
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students identify their own abilities and interests as possible guides to career choice.
- CDOS Standard 1 Career Development: Students will be knowledgeable about the world of work, explore career options, and relate personal skills, aptitudes, and abilities to future career decisions.
- National FCS 3.0: 8.1.1 Explain what people in food production and food service careers do.
- National FCS 3.0: 8.1.3 Sum up the education and training that food production and service career paths require and where they lead.
- National FCS 3.0: 9.1.3 Sum up the education and training that food science, dietetics, and nutrition career paths require.
- National FCS 3.0: 1.2.1 Examine possible careers to find what each one requires and offers in knowledge, skills, attitudes, and openings.

Note: Area 14 (Nutrition and Wellness) has no career competency in the 3.0 text, so the food career competencies are cited from Areas 8 and 9, the same substitution made in Lesson 1.22.

Enduring understanding and essential question

Big idea: Every bite of the meal this class cooked passed through a chain of jobs: growing, processing, inspecting, moving, buying, cooking, serving. Some of those jobs start at sixteen with a two-day certificate. Some take a doctorate. They are all in this chain, and several of them are ten minutes from here. **Essential question:** Which job in the food chain would I actually want, and what is the shortest honest path to it?

Objectives

By the end of this lesson, students will be able to: 1. (Low, Day 1) Given thirteen career cards, place each one on the farm-to-table chain (grow, process, inspect, move, sell, cook, serve, study) and sort them by the education needed. 2. (Mid, Day 1) Compare two careers on a day in the life, the training path, and where the job exists on Long Island, and state which one better fits the student's own interests with a reason. 3. (High, Day 2) On the Unit 6 test, apply all four topics of the unit (where food comes from; preparation methods and science; the Feed the Class system; careers) to twenty items including scaling math, unit price math, a scenario, and a written response.

Vocabulary

Tier 2 (general academic): pathway, credential, suitability, qualification Tier 3 (FACS): ServSafe, apprenticeship, certificate, associate degree, food scientist, food safety inspector, logistics, buyer, dietitian, agricultural engineer. See the unit vocabulary list for Turkish, Portuguese, and Spanish.

Materials and setup

- Handout, one set per pair plus one per student: Handout 06.18 - Food Systems Careers Cards.md (thirteen career cards to cut apart, the farm-to-table chain mat, the education sort strip, the “would I like this” self-check, and the Long Island jobs page)
- Assessment, one per student, Day 2: Assessment 06.3 - Unit 6 Test.md (student version; the teacher key is in the same file after the “Teacher key” heading and is not printed for students)
- Rubric 06 - Feed the Class.md, scored, returned at the start of Day 2
- The review game materials, Day 2: four sets of the review question cards on page 6 of Handout 06.18, four small whiteboards or four half sheets per team, a marker per team
- Pencils and a calculator per student (allowed on the two math items)
- Slides: Slides 06.18 - Careers in Food Systems and Unit 6 Review and Test (outline in 05 Materials; the site renders it as a deck; Day 1 and Day 2 are labeled)
- Setup notes: cut the card sets the day before Day 1. Print the test single sided so the math items have room to work. Score Rubric 06 the night before Day 2 so it comes back before the test. Leave the class master flowchart, the four station posters, and the two waste numbers on the wall for the review game. No food is handled either day.

Pre-assessment

Three things. The career self-check from Lesson 1.22 is in every student's FACS folder; pull it out at the start of Day 1 and have students read what they wrote in the fall. Unit 2's interest code from Lesson 2.2 is in the same folder. And the do now on Day 1 asks who has a relative or neighbor in any part of the food chain, which in most Long Island classes is most of the room.

Do now and hook

Day 1 (Time: 4 min). On the board: "Name one job somebody had to do before the food we cooked last week reached this room. Not a cook. Somebody earlier in the chain."

Debrief: take six answers and write them along a chain drawn across the board, left to right. Then the hook: "There are thirteen jobs on the cards in front of you. Two of them you can start at sixteen with a certificate that takes two days. One of them exists on a farm twenty-five miles from here. And one of them is why the prices on that circular are what they are."

Day 2 (Time: 4 min). Hand back the scored Rubric 06 as students walk in. On the board: "Read your rubric. Write one line: the criterion I scored highest on and the one thing I would change if we ran this project again."

Debrief: take three answers, no names on scores, then say what happens today: fifteen minutes of review as a team game, then the test in the last twenty minutes.

Procedure

Day 1: careers across the whole chain

Step	Teacher will	Students will	Time
1. Direct instruction: the chain has jobs at every link	Draw the eight links across the slide: grow, process, inspect, move, sell, cook, serve, study. Name one job per link out loud and say the one fact that makes it real (the inspector has legal authority to close a kitchen; the logistics coordinator decides whether the truck comes Tuesday or Thursday; the buyer is why the circular	Copy the eight links. Name the link for each of the six jobs from the do now.	8 min

	says what it says). Say the honest thing about pay and hours: kitchen jobs start low and the hours are nights and weekends; the science and inspection jobs pay more and take a degree; the sales and logistics jobs are the ones most students have never heard of and there are a lot of them.		
2. Learning activity: the chain mat and the education sort	Pairs place all thirteen cards on the chain mat, then re-sort the same thirteen by education needed into four columns: no degree needed (training on the job or a short certificate), certificate or apprenticeship, two-year degree, four-year degree or more. Circulate. The two cards that move the most are the caterer (any column, depending on the business) and the baker (on the job, an apprenticeship, or a culinary certificate).	Sort twice. Write one job per education column on the strip. Mark the two cards that were hardest to place and say why.	11 min
3. Direct instruction: the training paths, and ServSafe as a real credential	Walk the five paths on one slide: on-the-job training; a short food safety certificate; a	Write the five paths. Write next to each the earliest age or grade a person could start it.	8 min

	registered apprenticeship, which is paid work plus classroom hours; a certificate or two-year degree at a community college; a four-year degree or more. Then name ServSafe specifically: it is the food safety program of the National Restaurant Association's training arm, the Food Handler level is a short course and exam, the Food Protection Manager level is the one many restaurants require of a supervisor, and high school CTE programs can and do offer it, so a student can walk out of high school with it. Say what it is worth: it is the difference between an application that gets a call and one that does not. <i>[Sal: confirm which ServSafe level, if any, your school can offer and what a proctored exam costs; also confirm the county health department's current food handler requirement.]</i>		
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4. Practice: the “would I like this” self-check	Each student picks two cards, one they would consider and one they would not, and fills the self-check: the part of the day I would like, the part I would not, the training I am willing to do, and one sentence in the form “I would or would not want ___ because ___.” Then read the Long Island jobs page and circle any job that exists within driving distance.	Fill the self-check individually. Circle the local jobs. Compare to the Lesson 1.22 self-check from the fall and write one line on whether the answer changed.	7 min
5. Share and close	Take one sentence from four students, one per corner of the education sort. Ask the closing question: “Which link in the chain has the most jobs, and which one has the jobs nobody applies for?”	Four students read one sentence. Everyone writes the answer to the closing question on the self-check.	2 min

Day 1: $4 + 8 + 11 + 8 + 7 + 2 = 40$. Closure is folded into Step 5.

Day 2: the review game and the Unit 6 test

Step	Teacher will	Students will	Time
1. Rubric return and the do now	Hand back the scored Rubric 06 at the door. Take three answers to the do now. Say in one sentence what the class did well as a group and one thing to carry into Unit 7.	Read the rubric. Write the one-line answer.	4 min
2. Review game:	Run the game off	Answer on the team	15 min

Four Links	page 6 of the handout. Four teams, the same station teams. Twenty-four cards in four piles by topic (where food comes from; methods and science; the project system; careers). Read a card; every team writes an answer on the board and holds it up on the cue; a correct answer is a point, and a correct answer with a reason is two. Use the wall: the master flowchart, the four station posters, and the two waste numbers are all fair game and all visible. Keep score on the board.	board. Hold up on the cue. Give the reason when asked for the second point.	
3. Unit 6 test	Hand out the test. Read the directions once. Calculators out for items 16 and 17 only. Twenty minutes on the visible timer. Collect at the bell.	Take the test.	20 min
4. Closure	Collect. One word at the door.	Say the one word.	1 min

Day 2: $4 + 15 + 20 + 1 = 40$. The test runs in the last twenty minutes, as the calendar says. Never shorten it; if the review game runs long, stop it mid-round.

Questions to ask

Monitor understanding (Day 1): - Which link does a food safety inspector work at? Could they work at more than one? - Which of these thirteen needs the most school? Which needs none past high school? - What is the difference between a food scientist and a dietitian?

Who works with people and who works with samples? - What does a buyer for a supermarket actually do all day, and how would you find out? - Which of these jobs exist within twenty-five miles of here? Point to them on the Long Island page.

Deepen learning: - Two jobs on these cards did not exist fifty years ago. Which ones, and what changed? (Food photographer, and the supply chain and logistics coordinator in anything like its current form. Phones and cameras made one; national distribution and software made the other.) - The inspector can shut a kitchen down. Why does the state give one person that power, and who is it protecting? - Kitchen work pays low and runs nights and weekends. Why do people still do it? Is that a good enough reason? - If a farm near here cannot find enough workers, what happens to the price on the circular? Trace it through the chain. - Which of these thirteen jobs would a computer have the hardest time doing, and why?

Check for understanding

Day 1: during Step 2, a clipboard yes or no per pair on “the chain mat placement is defensible” and “the education sort matches the key on the handout for at least ten of thirteen.” During Step 4, read two self-checks over a shoulder and look for a reason, not a preference. Day 2: the review game is the check, and it is public and immediate, one held-up board per team per card. The test is the summative check for the whole unit.

Closure

Day 1 (Time: 2 min, inside Step 5). Four students read their “I would or would not want ___ because ___” sentence. Everyone answers the closing question on the sheet. The self-check goes in the FACS folder next to the Lesson 1.22 one; Unit 7 uses both.

Day 2 (Time: 1 min). Collect the tests. Each student says one word on the way out: the best thing about this unit, in one word. Write the words on the board for the next section to see. Say the last thing: “Next unit you build a resume and pitch a business. Two of the thirteen cards from yesterday come back.”

Differentiation and supports

- ELL: every career card has a picture of the person at work, a one-line “a day in this job” in short sentences, and a symbol for the education level; the chain mat is a picture of the eight links; the self-check has a sentence frame in English, Turkish, Portuguese, and Spanish; the test is read aloud in a small group for any student with that support, with the item wording unchanged; ServSafe, apprenticeship, certificate, and degree are in Vocabulary 06.md in all four languages. *[Sal: a translated test is the ELL coordinator's decision; the file is plain enough to translate.]*
- IEP and 504 (general): extended time on the test (finish at the start of Unit 7 Lesson 7.1 or in a support period); the test may be taken in a separate location; calculators for anyone who wants one, not just on items 16 and 17; the card sorts are physical and can be done on a desk with the cards handed over rather than reached for; the self-check may be dictated; the written-response item accepts a labeled drawing or a bulleted answer; the review game answer may be spoken to a teammate who writes it.

- Grade 6 support: eight career cards instead of thirteen (farmer, processing technician, truck driver or logistics coordinator, grocery store manager, line cook, baker, dietitian, food safety inspector). The test omits items 19 and 20 and is scored out of 28. Or keep all twenty items and give the extra time.
- Grade 8 stretch: the pair adds a fourteenth card for a food-chain job not on the list (aquaculture farmer, food truck owner, packaging designer, agricultural drone operator, school food service director) with the link, the education, a day in the life, and where it exists locally. On the test, the written-response item asks for two reasons and one counterargument.
- UDL checkpoint used: multiple means of representation (the chain as a drawn line, the careers as picture cards, the paths as a five-step ladder) and multiple means of engagement (the self-check is about the student's own future and has no right answer, and the review game is a team competition with the room's own work on the walls as the reference).
- No-kitchen alternative: not needed; paper only both days.

Allergy line

No food is handled either day. The Long Island jobs page names local farms and food businesses; if the class is invited to visit one, the allergy list travels with the class and the trip is cleared with the nurse first.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. Place thirteen careers on the chain and sort by education	FSP 6. a); CCO 4. a); National FCS 8.1.1, 8.1.3	Chain mat and education sort, Day 1 Step 2	Formative; sorts checked against the key on the handout
2. Compare two careers and state a personal fit with a reason	FSP 6. b), 6. c); CCO 7. c); SUS 4. a); Std 3 PI E; CDOS 1; National FCS 1.2.1, 9.1.3	Self-check and the Long Island page, Day 1 Step 4	Formative; scored complete, partial, or missing for daily work; filed for Unit 7
3. Apply all four unit topics on the test	FSP 1. c), 1. d), 2. a), 2. b), 3. a), 4. a), 4. d), 5. a), 5. c), 5. d), 6. a); SUS 1. a), 2. h); FCL 4. g); HSW 4. c)	Unit 6 test, Day 2 Step 3	Summative; Assessment 06.3 - Unit 6 Test.md, 20 items, 40 points, standard line per item in the key; topic assessments category

Topic 6.4 has no separate check; it is assessed inside the unit test, as the scope and sequence says.

Homework

None. The unit is done. Unit 7 starts the next day with the resume, and the self-check from today is one of the two documents Lesson 7.2 builds from.

Connections

Inside the building: the school counselor. FACS gives the counselor thirteen food-chain careers with education paths and a completed student self-check for the file, plus the Unit 2 interest code to read next to it; the counselor gives FACS the NYS Career Zone login routine, your school's high school course sheet, and whether any CTE certificate, including ServSafe, is available in the building or through a BOCES program. Community: four partners, any one of which turns a card into a person, for ten minutes in person or on video. A supermarket manager (the buyer and store manager cards). A Cornell Cooperative Extension of Suffolk County educator (the farmer, the agricultural engineer, and the food safety cards). A local farm or farmers market vendor on the North Fork or at the nearest market (the grower cards). A caterer or restaurant owner (the chef, line cook, and owner cards). *[Sal: name the four and write the ask in one email each; Cooperative Extension runs school programs already and is the easiest yes.]*

Sources

- NYS Middle Level CTE FACS Content Module 06, Food Systems and Production (nyctecenter.org, June 2018), lines 6. a), 6. b), and 6. c). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 01, Career and Community Opportunities, lines 4. a) and 7. c); Theme Module 06, Sustainability, line 4. a). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6. *[Sal: confirm against the NYSED PDF.]*
- CDOS Intermediate Standard 1, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 1.2.1, 8.1.1, 8.1.3, and 9.1.3, per 02 Standards/STANDARDS SOURCES - web research.md.
- Pay ranges and job outlook lines on the career cards are marked *[update]* and should be filled from the U.S. Bureau of Labor Statistics Occupational Outlook Handbook (bls.gov/ooh) and New York State Department of Labor wage data for the Long Island region. The registered dietitian credential requires a graduate degree to sit for the exam as of 2024; confirm with the Commission on Dietetic Registration. ServSafe is administered through the National Restaurant Association's training arm; confirm the current levels, exam fee, and proctoring rules at servsafe.com. Registered apprenticeship information for New York comes from the NYS Department of Labor apprenticeship office. *[Sal: confirm all four before the site goes public.]*

- Sal's own materials: the careers-day structure follows Lesson 1.22 in this course, and the Long Island jobs page draws on his own King Kullen price research (King Kullen is a Long Island chain and is widely credited as the first American supermarket, opened in Queens in 1930, which is a one-line hook worth using) and his hotel food and beverage work on Fire Island.

Teacher notes

- Day 1's best five minutes are the two cards that will not stay put. Let the caterer and the baker start an argument about which education column they belong in. The honest answer is "it depends on the business," and that is the lesson.
- Do not oversell kitchen work. Students who watch cooking shows think chef means fame. The cards say nights, weekends, standing, and a starting wage. Let the numbers argue, the same way Lesson 1.22 does with the pay sort.
- The two cards students have never heard of are the supply chain and logistics coordinator and the processing technician, and those two links have the most openings. Spend a full minute on each.
- ServSafe is the single most useful sentence in this lesson for a student who is not going to a four-year college. Say it plainly: a certificate you can earn in high school that a restaurant actually asks for.
- Day 2: return the rubric before the game, not after the test. A student who finds out they scored well plays the game harder and takes the test calmer.
- The review game uses the walls on purpose. Students who look up at their own flowchart to answer a question are using their own work as a reference, which is what the test lets them do from memory.
- *[Sal: your own path is one of the cards in a way. Corporate Innovation, an MBA, Oracle, hospitality, then this room. Tell the class in two sentences that the path is not a straight line. It is the most useful thing a fourteen-year-old can hear on a careers day.]*
- *[Sal: if a Cooperative Extension educator or a North Fork farm will take a class visit or a video call in late May, put it here and drop the review game to ten minutes. A real grower beats the cards.]*